Measuring Fairness through Calibration in Insurance Scoring Algorithms

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Motivation: Using Probabilistic Classifiers

- Several applications of (binary) classification require **probabilistic classifiers**.
  - Rather than predicting labels in $\{0, 1\}$, they output the estimated likelihood of the positive class given the features, *i.e.*, **confidence scores** $\in [0, 1]$.
- For example, in **insurance pricing**:
  - Estimation of **mortality scores**, $\mathbb{P}(Y = 1 \mid \boldsymbol{X} = \boldsymbol{x})$, where $Y \in \{0, 1\}$ indicates the occurrence of death (Sauce et al., 2023).
  - The **probability that an insured has an accident** within the year is estimated and directly used to calculate the pure premium.
- Using a **GLM**, we can estimate

$$\mathbb{E}[Y \mid \boldsymbol{X}] = \mathbb{P}(Y = 1 \mid \boldsymbol{X}) = \overbrace{\frac{\exp[\boldsymbol{X}^\top \boldsymbol{\beta}]}{1 + \exp[\boldsymbol{X}^\top \boldsymbol{\beta}]}}^{\text{“probabilistic scores”}} = s(\boldsymbol{X})$$

Motivation: Machine Learning for Insurance Pricing

- Machine learning **models** are increasingly used by insurers to provide more **personalized premiums** (NAAC, 2022) or to handle **high-dimensional data** (Gao et al., 2018).
- Using proper loss functions, such as the log-loss, we can decompose the divergence d_ϕ between the true probability $\mathbb{P}(Y \mid \boldsymbol{X})$ and the estimated scores $s(\boldsymbol{X})$ (Gneiting and Raftery, 2007; Kull and Flach, 2015)

$$\mathbb{E}\big[d_\phi(s(\boldsymbol{X}), \mathbb{P}_{Y|\boldsymbol{X}})\big] = \textbf{Calibration Loss} + \text{Grouping Loss}.$$

  - To accurately interpret scores as probabilities of occurrence, we need **Calibration Loss** $= 0$.
- "*Guo et al. (2017) have shown that* **modern neural networks** *(NN) are* **poorly calibrated** *and over-confident despite having better performance*."

This Talk

In brief,

1. Using a machine learning binary classifier, we estimate **confidence scores as probabilities of occurrence** of a binary event.
2. The confidence scores do not necessarily have the same **characteristics across groups** defined by a sensitive attribute.

   How can we correct the <u>unfairness</u> in the confidence scores of a binary classifier?

3. In the literature,
   - **Calibration metrics** calculated within sensitive groups,
   - **Mitigation techniques** to achieve fairness with respect to those calibration metrics.

Table of Contents

What the remainder of the talk is about:

1. Presentation of the notion of **calibration** to assess the **confidence scores** of a binary classifier.
2. Review of methods to **measure and visualize calibration** for a binary classifier.
3. Review of **metrics to evaluate unfairness** based on calibration errors across sensitive groups.
4. Application of two **unfairness mitigation techniques** based on calibration correction to a life insurance dataset.

Framework and Notations

- $A \in \mathcal{A} = \{0, 1\}$ denotes a "protected attribute" (*e.g.*, sex or race),
- $\boldsymbol{X} \in \mathcal{X} = \mathbb{R}^d$ are considered "non-protected" explanatory variables,
- $Y \in \mathcal{Y} = \{0, 1\}$ is the binary outcome variable we aim to predict,
- $s : \mathcal{Z} \to [0, 1]$ is a **binary probabilistic classifier**, where $\mathcal{Z} = \mathcal{X} \times \mathcal{A}$,

$$s(\boldsymbol{X}, A) := \widehat{\mathbb{E}}[Y \mid \boldsymbol{X}, A] = \widehat{\mathbb{P}}(Y = 1 \mid \boldsymbol{X}, A).$$

  - We denote the **Bayes-optimal score**

$$\mu(\boldsymbol{X}, A) := \mathbb{E}[Y \mid \boldsymbol{X}, A] = \mathbb{P}(Y = 1 \mid \boldsymbol{X}, A).$$

Framework: Calibration can be assessed for all observations in a dataset as a **measure of performance** for a binary classifier or between groups as a **measure of unfairness**, *e.g.*, for $A = 0$ and $A = 1$.

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Insurance Pricing Criteria: "Overall" Calibration

Global balance:

$$\mathbb{E}\big[Y - s(\boldsymbol{Z})\big] = \mathbb{E}\big[\mu(\boldsymbol{Z}) - s(\boldsymbol{Z})\big] = 0.$$

- Economically, the **portfolio** is self-financing (for Y).

Calibration (Dawid, 1982) (or auto-calibration (Denuit et al., 2021)) "*among insureds to whom a calibrated mortality-risk classifier assigns a probability of 0.6, 60% will actually die within the year*"

$$\mathbb{E}\big[Y - s(\boldsymbol{Z}) \mid s(\boldsymbol{Z})\big] \overset{\text{a.s.}}{=} \mathbb{E}\big[\mu(\boldsymbol{Z}) - s(\boldsymbol{Z}) \mid s(\boldsymbol{Z})\big] \overset{\text{a.s.}}{=} 0.$$

or equivalently, $\mathbb{E}\big[Y \mid s(\boldsymbol{Z}) = p\big] = p$, for all $p \in \mathrm{Supp}\,(s(\boldsymbol{Z}))$.

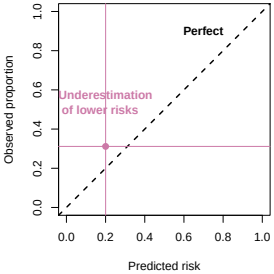
- Economically, **price-based subgroups** $s(\boldsymbol{Z})$ are self-financing (for Y).

Visualizing Calibration

To evaluate the **calibration** of a model s, we can plot its **calibration curve**:

$$\forall p \in [0,1], \quad g : \begin{cases} [0,1] \to [0,1] \\ p \mapsto g(p) := \mathbb{E}[Y \mid s(\boldsymbol{Z}) = p] \end{cases} .$$

- For a confidence score of 0.2, 25 out of 100 in this group are found to have the disease: the risk is **underestimated**.



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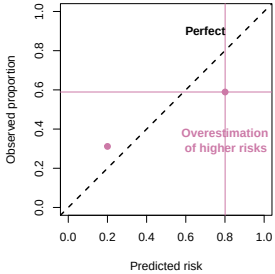
- For a confidence score of 0.8, 60 out of 100 in this group are found to have the disease: the risk is **overestimated**.



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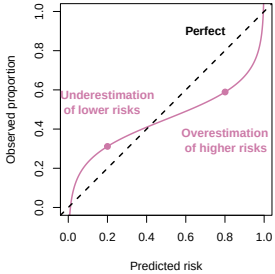


Assessing Calibration

Based on the **distance between the calibration curve g and the bisector**, the **calibration error** of a probabilistic classifier s can be estimated using

- The **Expected Calibration Error** (ECE) (Pakdaman Naeini et al., 2015): using a binning scheme based on score quantiles;
- **Isotonic regression**, which provides an optimal binning scheme on s (Dimitriadis et al., 2021);
- The **Integrated Calibration Index** (ICI) (Austin and Steyerberg, 2019): estimating the calibration curve via local regression (Fernandes Machado et al., 2024).

Framework: When those calibration metrics are calculated over the entire dataset, they serve as a **measure of performance** for a probabilistic binary classifier.

Calibration of Insurance Scoring Algorithms

Algorithmic Fairness: The Sufficiency Criterion through Calibration

Unfairness Mitigation Algorithms for a Life Insurance Dataset

Overview of Group-Fairness Criteria

- The notion of fairness for groups based on A (*e.g.*, **male** *vs* **female**) differs from *actuarial fairness*, "*requiring premiums to be set proportional to an individual's risk*." (Baumann and Loi, 2023; Arrow, 1963)
- We can use different **group fairness metrics** to evaluate the unfairness of a model s w.r.t. A (Barocas et al., 2023; Charpentier, 2024).
  1. The **Independence** criterion requiring $s(\boldsymbol{Z}) \perp A$, e.g., Demographic Parity (Fernandes Machado et al., 2025);
  2. The **Separation** criterion requiring $s(\boldsymbol{Z}) \perp A \mid Y$, e.g., Equalized Odds;
  3. The **Sufficiency** criterion requiring $Y \perp A \mid s(\boldsymbol{Z})$, e.g., **similar calibration across sensitive groups**.
- Due to **impossibility theorems**, these criteria cannot be mathematically achieved simultaneously (Kleinberg et al., 2016), unless very strict assumptions are met.

Sufficiency Criterion in Insurance Pricing

Why use the **Sufficiency Criterion** (similar calibration errors across sensitive groups) *rather than Independence or Separation to evaluate unfairness in an insurance context?*

- For insurance pricing, compared to the traditional predictive modeling framework, **the true risk of policyholders cannot be measured** at the individual level, which may render some fairness criteria irrelevant (Baumann and Loi, 2023).
- The Sufficiency criterion is widely used in **healthcare** (Obermeyer et al., 2019; Liu et al., 2025) and should also be the case for **life insurance pricing** models.
- In practice, the **cost of fairness** (deviating from *actuarial fairness*) required to achieve Sufficiency is lower than that for Independence or Separation (Liu et al., 2019).

Calibration-based Unfairness Metrics

- **Calibration parity** (Chouldechova, 2017):

$$\forall p \in [0,1],\ \ \mathbb{P}\big(Y = 1 | s(\boldsymbol{X}, A) = p, A = 0\big) = \mathbb{P}\big(Y = 1 | s(\boldsymbol{X}, A) = p, A = 1\big).$$

- **Group-wise calibration** (Kleinberg et al., 2016):

$$\forall p \in [0,1],\ \ \mathbb{P}\big(Y = 1 | s(\boldsymbol{X}, A) = p, A = 0\big) = \mathbb{P}\big(Y = 1 | s(\boldsymbol{X}, A) = p, A = 1\big) = p.$$

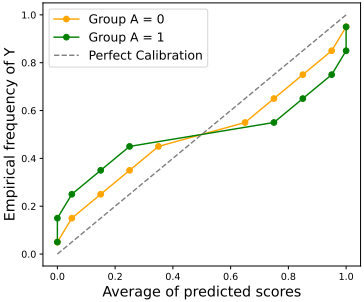
- **Multicalibration** (Hebert-Johnson et al., 2018): this notion is **stricter than group-wise calibration**, as it requires predictions to be calibrated for every subpopulation within a family of sets based on $\mathcal{X}$, where all sensitive groups are included in $\mathcal{X}$ (potentially overlapping groups).

Example: *Calibration parity*

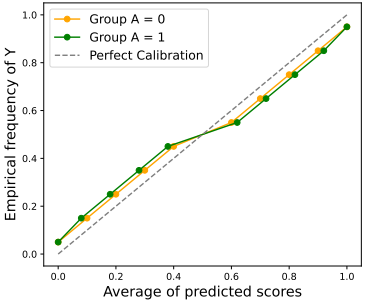
Consider a model s estimating the **probability that a policyholder experiences an accident** within a year. We evaluate the unfairness of s by requiring the **same calibration errors** for both sensitive groups $A = 0$ and $A = 1$:

$$\forall p \in [0,1], \;\; \mathbb{P}\big(Y = 1 | s(\boldsymbol{X}, A) = p, A = 0\big) = \mathbb{P}\big(Y = 1 | s(\boldsymbol{X}, A) = p, A = 1\big)$$

Unfair model s



Fair model $\tilde{s}$

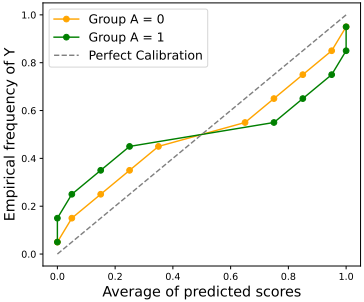


Example: *Group-wise calibration*

Consider a model s estimating the **probability of developing a disease** for different insureds. We evaluate the unfairness of s by requiring **no calibration errors** for both sensitive groups $A = 0$ and $A = 1$:

$$\forall p \in [0,1], \;\; \mathbb{P}\big(Y = 1 | s(\boldsymbol{X}, A) = p, A = 0\big) = \mathbb{P}\big(Y = 1 | s(\boldsymbol{X}, A) = p, A = 1\big) = p.$$

Unfair model s



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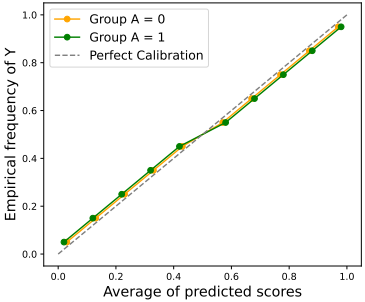

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Overview of Unfairness Mitigation Techniques

To correct for unfairness in model predictions, mitigation approaches can be applied at different stages of the machine learning pipeline:

1. **Pre-processing**: Modify training data to reduce bias **before model learning**.
2. **In-processing**: Incorporate fairness constraints **into the learning** algorithm.
3. **Post-processing**: Adjust model predictions **after model training**.

- In this study, we will consider two **post-processing mitigation approaches** to achieve **group-wise calibration** under the group fairness criterion of Sufficiency.

Post-processing Methods for Sufficiency

- Hebert-Johnson et al. (2018) propose an **iterative boosting algorithm** (the **HKRR** method): the algorithm works by iteratively identifying and removing calibration violations within subgroups until convergence.
- Haghtalab et al. (2023) applies game-theoretic concepts in the context of **multi-objective learning** (the **HJZ** method), where learning guarantees must hold simultaneously across a set of distributions and loss functions, defined here as calibration within each subgroup.

SEER dataset

- Prediction of **1-year mortality scores** for individuals diagnosed with skin cancer (`https://seer.cancer.gov`) in the United States (Sauce et al., 2023).
- The dataset contains $n = 547{,}878$ individuals from 2004 to 2018.
- We have the following explanatory variables, $(\boldsymbol{X}, A)$:
  - **Demographic data:** duration, age, gender, ethnicity, marital status.
  - **Medical data:** time since tumor onset, mitotic rate, primary tumor site, spread, tumor size, laterality, ulceration, metastases.
- The **sensitive attribute** considered here is **gender** (A) with two categories: **female** and **male**.

Group-wise Calibration Metric

- The calibration-based unfairness metric $\mathcal{U}$ of model s is calculated, based on the ECE, as

$$\mathcal{U}(s) := \max\left\{\mathrm{ECE}(s \mid A = \textbf{female}), \mathrm{ECE}(s \mid A = \textbf{male})\right\},$$

where the **ECE** for a model s, considering all observations, is defined as (Pakdaman Naeini et al., 2015)

$$\mathrm{ECE}(s) := \sum_{b \in [B]} \frac{n_b}{n} \cdot |\mathrm{freq}_b - \mathrm{conf}_b|$$

where the bins $1, \ldots, B$ are defined according to the quantiles of the predicted scores, and for n realizations $(\boldsymbol{x}_i, a_i, y_i)$ of $(\boldsymbol{X}, A, Y)$:

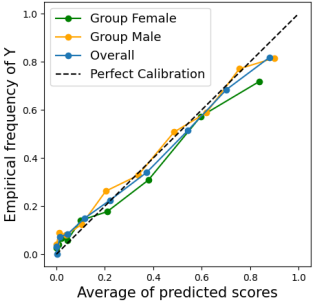
- $\mathrm{conf}_b = \frac{1}{n_b} \sum_{i \in b} s(\boldsymbol{x}_i, a_i)$,
- $\mathrm{freq}_b = \frac{1}{n_b} \sum_{i \in b} y_i$.

Application of Unfairness Mitigation Methods

We apply the two correction algorithms (HKRR and HJZ) on the SEER dataset.

- See the Appendix for more details on the procedure in practice.

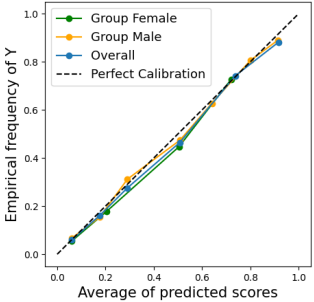
Initial model s



$\mathrm{ECE}(s \mid A = \textbf{female})$: 0.042

$\mathrm{ECE}(s \mid A = \textbf{male})$: 0.035

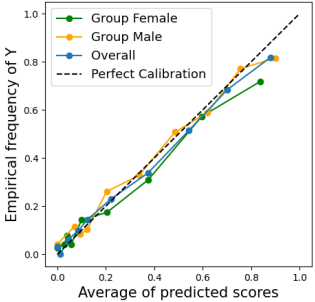
Fair model with HKRR



$\mathrm{ECE}(s \mid A = \textbf{female})$: 0.031

$\mathrm{ECE}(s \mid A = \textbf{male})$: 0.023

Fair model with HJZ



$\mathrm{ECE}(s \mid A = \textbf{female})$: 0.040

$\mathrm{ECE}(s \mid A = \textbf{male})$: 0.027

Future Work: Heterogeneity of Scores in Sensitive Groups

- Consider the case of a probabilistic classifier that estimates the probability of a "**positive outcome**" Y, such as credit approval or hospital admission.
- Due to historical injustices, confidence **score ranges may vary between sensitive groups**, with disadvantaged groups having lower scores.
- La Cava et al. (2023) introduced **Proportional Multicalibration** as a first step toward addressing this: "*By relating sufficiency to multicalibration (MC), we describe a shortcoming of MC that can occur when the outcome probabilities are strongly tied to group membership,* **MC does not provide equal protection to groups with different risk profiles**."

Conclusion

- **Calibration** is one type of metric used to assess the **performance of a binary probabilistic classifier**.
- When **evaluated across sensitive groups**, calibration can serve as an **unfairness metric**: it is widely used in healthcare and relevant in insurance.
- Future work: due to **variability in score ranges across sensitive groups**, mitigation algorithms may still favor majority groups.

References I

Arrow, K. J. (1963). Uncertainty and the welfare economics of medical care. *The American Economic Review*, 53(5):941–973.

Austin, P. C. and Steyerberg, E. W. (2019). The Integrated Calibration Index (ICI) and related metrics for quantifying the calibration of logistic regression models. *Statistics in Medicine*, 38(21):4051–4065.

Barocas, S., Hardt, M., and Narayanan, A. (2023). *Fairness and Machine Learning: Limitations and Opportunities*. Adaptive Computation and Machine Learning series. MIT Press.

Baumann, J. and Loi, M. (2023). Fairness and risk: An ethical argument for a group fairness definition insurers can use. *Philosophy & Technology*, 36(3):45.

Charpentier, A. (2024). *Insurance, biases, discrimination and fairness*. Springer.

Chouldechova, A. (2017). Fair prediction with disparate impact: A study of bias in recidivism prediction instruments. *Big Data*, 5(2):153–163. PMID: 28632438.

Dawid, A. P. (1982). The well-calibrated bayesian. *Journal of the American Statistical Association*, 77(379):605–610.

Denuit, M., Charpentier, A., and Trufin, J. (2021). Autocalibration and tweedie-dominance for insurance pricing with machine learning. *Insurance: Mathematics and Economics*, 101:485–497.

Dimitriadis, T., Gneiting, T., and Jordan, A. I. (2021). Stable reliability diagrams for probabilistic classifiers. *Proceedings of the National Academy of Sciences*, 118(8):e2016191118.

Fernandes Machado, A., Charpentier, A., Flachaire, E., Gallic, E., and Hu, F. (2024). From uncertainty to precision: Enhancing binary classifier performance through calibration.

Fernandes Machado, A., Grondin, S., Ratz, P., Charpentier, A., and Hu, F. (2025). Equipy: Sequential fairness using optimal transport in python.

Gao, G., Meng, S., and Wüthrich, M. V. (2018). Claims frequency modeling using telematics car driving data. *Scandinavian Actuarial Journal*.

Gneiting, T. and Raftery, A. E. (2007). Strictly proper scoring rules, prediction, and estimation. *Journal of the American Statistical Association*, 102(477):359–378.

Guo, C., Pleiss, G., Sun, Y., and Weinberger, K. Q. (2017). On calibration of modern neural networks. In *International conference on machine learning*, pages 1321–1330. PMLR.

References II

Haghtalab, N., Jordan, M., and Zhao, E. (2023). A unifying perspective on multi-calibration: Game dynamics for multi-objective learning. In Oh, A., Naumann, T., Globerson, A., Saenko, K., Hardt, M., and Levine, S., editors, *Advances in Neural Information Processing Systems*, volume 36, pages 72464–72506. Curran Associates, Inc.

Hansen, D., Devic, S., Nakkiran, P., and Sharan, V. (2024). When is multicalibration post-processing necessary? In *The Thirty-eighth Annual Conference on Neural Information Processing Systems*.

Hebert-Johnson, U., Kim, M., Reingold, O., and Rothblum, G. (2018). Multicalibration: Calibration for the (Computationally-identifiable) masses. In Dy, J. and Krause, A., editors, *Proceedings of the 35th International Conference on Machine Learning*, volume 80 of *Proceedings of Machine Learning Research*, pages 1939–1948. PMLR.

Kleinberg, J., Mullainathan, S., and Raghavan, M. (2016). Inherent trade-offs in the fair determination of risk scores.

Kull, M. and Flach, P. (2015). Novel decompositions of proper scoring rules for classification: Score adjustment as precursor to calibration. In *Machine Learning and Knowledge Discovery in Databases: European Conference, ECML PKDD 2015, Porto, Portugal, September 7-11, 2015, Proceedings, Part I 15*, pages 68–85. Springer.

La Cava, W. G., Lett, E., and Wan, G. (2023). Fair admission risk prediction with proportional multicalibration. In Mortazavi, B. J., Sarker, T., Beam, A., and Ho, J. C., editors, *Proceedings of the Conference on Health, Inference, and Learning*, volume 209 of *Proceedings of Machine Learning Research*, pages 350–378. PMLR.

Liu, L. T., Simchowitz, M., and Hardt, M. (2019). The implicit fairness criterion of unconstrained learning. In Chaudhuri, K. and Salakhutdinov, R., editors, *Proceedings of the 36th International Conference on Machine Learning*, volume 97 of *Proceedings of Machine Learning Research*, pages 4051–4060. PMLR.

Liu, M., Ning, Y., Wang, H., Hong, C., Engelhard, M., Bitterman, D. S., Cava, W. G. L., and Liu, N. (2025). Equitable survival prediction: A fairness-aware survival modeling (fasm) approach.

NAAC (2022). Appendix b-trees – information elements and guidance for a regulator to meet best practices' objectives (when reviewing tree-based models).

Obermeyer, Z., Powers, B., Vogeli, C., and Mullainathan, S. (2019). Dissecting racial bias in an algorithm used to manage the health of populations. *Science*, 366(6464):447–453.

Pakdaman Naeini, M., Cooper, G., and Hauskrecht, M. (2015). Obtaining well calibrated probabilities using bayesian binning. *Proceedings of the AAAI Conference on Artificial Intelligence*, 29(1):2901–2907.

Sauce, M., Chancel, A., and Ly, A. (2023). Ai and ethics in insurance: a new solution to mitigate proxy discrimination in risk modeling.

The Cost of Fairness

Is there a cost in predictive performance when requiring Sufficiency?

- Here are some useful properties on the Bayes optimal score:
  1. It is calibrated (overall).
  2. It satisfies the group-wise calibration criterion.
  3. It satisfies the calibration parity criterion.

Sufficiency and Optimal Score (1/2)

Let $\mu(\mathbf{X}, A) = \mathbb{E}[Y \mid \boldsymbol{X}, A] = \mathbb{P}(Y = 1 \mid \boldsymbol{X}, A)$ be the optimal score that we aim to estimate using a model s.

Proposition (Barocas et al., 2023; Liu et al., 2019)

$\mu(\boldsymbol{X}, A)$ satisfies the *group-wise calibration* criterion, i.e.,

$$\mathbb{E}\big[Y \mid \mu(\boldsymbol{X}, A), A\big] = \mu(\boldsymbol{X}, A).$$

Proof[1]:

$$\begin{aligned}
\mathbb{E}\big[Y \mid \mu(\boldsymbol{X}, A), A\big] &= \mathbb{E}\Big[\mathbb{E}\big[Y \mid \boldsymbol{X}, A\big] \mid \mu(\boldsymbol{X}, A), A\Big] \\
&= \mathbb{E}\big[\mu(\boldsymbol{X}, A) \mid \mu(\boldsymbol{X}, A), A\big] \\
&= \mu(\boldsymbol{X}, A)\,.
\end{aligned}$$

[1] The following equalities hold almost surely.

Sufficiency and Optimal Score (2/2)

Proposition (Barocas et al., 2023; Liu et al., 2019)

$\mu(\boldsymbol{X}, A)$ satisfies the *calibration parity* criterion, i.e.,

$$\mathbb{E}[Y \mid \mu(\boldsymbol{X}, A), A] = \mathbb{E}[Y \mid \mu(\boldsymbol{X}, A)].$$

Proof[2]:

From the previous proof,

$$\mathbb{E}\big[Y \mid \mu(\boldsymbol{X}, A), A\big] = \mu(\boldsymbol{X}, A).$$

Moreover,

$$\begin{aligned}\mathbb{E}\big[Y \mid \mu(\boldsymbol{X}, A)\big] &= \mathbb{E}\Big[\mathbb{E}\big[Y \mid \boldsymbol{X}, A\big] \,\Big|\, \mu(\boldsymbol{X}, A)\Big] \\ &= \mathbb{E}\big[\mu(\boldsymbol{X}, A) \mid \mu(\boldsymbol{X}, A), A\big] \\ &= \mu(\boldsymbol{X}, A)\,.\end{aligned}$$

Thus,

$$\mathbb{E}\big[Y \mid \mu(\boldsymbol{X}, A), A\big] = \mathbb{E}\big[Y \mid \mu(\boldsymbol{X}, A)\big].$$

[2] The following equalities hold almost surely.

Sufficiency and *Group-wise Calibration*

Let $s : \mathcal{X} \times \mathcal{S} \to \mathcal{Y}$.

Proposition (Barocas et al., 2023; Liu et al., 2019)

If s satisfies the group-wise calibration criterion, then s satisfies the *calibration parity* criterion.

Proof[3]:

Assume that

$$\mathbb{E}\big[Y \mid s(\boldsymbol{X}, A), S\big] = s(\boldsymbol{X}, A)$$

(*group-wise calibration* criterion).

We show that

$$\mathbb{E}\big[Y \mid s(\boldsymbol{X}, A), A\big] = \mathbb{E}\big[Y \mid s(\boldsymbol{X}, A)\big]$$

(*calibration parity* criterion).

Thus, it suffices to show that

$$\mathbb{E}\big[Y \mid s(\boldsymbol{X}, A)\big] = s(\boldsymbol{X}, A).$$

In Practice: Steps for the Mitigation Procedure

We can apply the **HKRR** and **HJZ** mitigation procedures using the approach, using the Python package `multicalibration` (Hansen et al., 2024):

1. Split the dataset into **three samples**: *train*, *calibration*, and *test*.
2. **Fit the predictive model** s on the *train* sample, using both $\boldsymbol{X}$ and A.
3. Apply the fitted model s on the *calibration* and *test* samples to predict the 1-year mortality scores.
4. **Fit the post-processing method**, denoted g_1 or g_2, on the *calibration* sample.
5. Apply the method g_1 or g_2 to obtain 1-year mortality scores calibrated by sensitive group on the *test* sample.

We will denote the two **fair predictors**: $\tilde{s}_1 = g_1 \circ s$ and $\tilde{s}_2 = g_2 \circ s$.